1. Suppose X and Y are two random variables. Please explain the meaning of $Cov(X,Y) < 0$, and write down the formula for $Cov(X,Y)$.

The **covariance** between two random variables *X* and *Y*, sometimes called the population covariance to emphasize that it concerns the relationship between two variables describing a population, is defined as the expected value of the product.

The covariance of two random variables X and Y is defined as:

$$Cov(X,Y) = E[X - E(X)][Y - E(Y)]$$

Clearly, $Cov(X,Y) = E(XY) - E(X)E(Y)$.

$$Cov(X,Y) = E[(X - E(X))(Y - E(Y))]$$

$$= E[XY - XE(Y) - E(X)Y + E(X)E(Y)]$$

$$= E(XY) - E(XE(Y)) - E(E(X)Y) + E[E(X)E(Y)]$$

$$= E(XY) - E(X)E(Y) - E(X)E(Y) + E(X)E(Y)$$

$$= E(XY) - E(X)E(Y)$$

If two random variables are associated (dependent), knowing the value of one (for example, X) will help to predict the likely value of the other (for example, Y). Covariance summarizes the strength of the linear association in a single number.

(b) Discussed during the lecture.

(c)

$$Cov(X,Y) = E[X - E(X)][Y - E(Y)]$$

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)} \times \sqrt{Var(Y)}}$$

$$= \frac{E[(X - E(X))(Y - E(Y))]}{sd(X)sd(Y)}$$

Correlation measures the strength of the linear (`straight-line') association between X and Y.

The further the correlation is from 0, the stronger is the linear association. The most extreme possible values of correlation are +1 and +1.

2. Suppose $E(X) = 2,\ E(Y) = 3,\ E(XY) = 4$. Calculate $Cov\left(\frac{2}{3}X, \frac{1}{3}Y\right)$

We know $Cov(X, Y) = E[X - E(X)][Y - E(Y)]$.

$$Cov\left(\frac{2}{3}X, \frac{1}{3}Y\right)$$

$$= E\left[\frac{2}{3}X - E\left(\frac{2}{3}X\right)\right]\left[\frac{1}{3}Y - E\left(\frac{1}{3}Y\right)\right]$$

$$= \frac{2}{9}[E(XY) - E(X)E(Y)]$$

$$= \frac{2}{9}(4 - 6)$$

$$= -\frac{4}{9}$$

3.

(a) $E(3X) = 30$, $E(3X - 20) = 10$, $E(4) = 4$ and $E\left(\frac{1}{2}X\right) = 5$

(b) $E(X + Y) = 30$ and $E(2X + 3Y) = 80$

Remarks:

Some Rules on Expectation Operator

Property E.1:
For any constant c, $E(c) = c$

Property E.2: If X and Y are random variables and a, b are constant, then

For any constants a and b, $E(aX + bY) = aE(X) + bE(Y)$

Property E.3:
If $\{a_1, a_2, \dots, a_p\}$ are constants and $\{X_1, X_2, \dots, X_p\}$are random variables, then
$E\left(a_1X_1 + a_2X_2 + a_3X_3 + \cdots + a_pX_p\right) = a_1E(X_1) + a_2E(X_2) + \cdots + a_pE\left(X_p\right)$

(c) $Var(2X) = 8$, $Var(4Y) = 16$ and $Var(X+Y) = 3$

(d) $Var(X+2Y) = 6$ and $Var(X-2Y) = 6$

Remarks:

Some Rules on Variance Operator

Def: $Var(X) \equiv E[X - E(X)]^2$

Property V.1: If c is a constant, then
$Var(c) = 0$

Property V.2: If X is random variable and a is constant, then
$Var(aX) = E[aX - E(aX)]^2 = a^2 Var(X)$

Property V.3: If X and Y are random variable, then
$Var(X \pm Y) = Var(X) + Var(Y) \pm 2Cov(X, Y)$

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(a) $E(X|X=1) = 1$

(b) $E(2X + 4Y|X=1) = E(2*1|X=1) + E(4Y|X=1) = 16$

(c) $E(Y|X=0) = 3.5$

(d) $Var(2Y + 3X|X=0) = Var(2Y|X=0) = 4Var(Y) = \frac{105}{9}$

Remarks

Some Rules on Conditional Expectation Operator

Property CE.1: $E[c(X)|X] = c(X)$ for any function $c(X)$.

Property CE.2: For functions $a(X)$ and $b(X)$, $E[a(X)Y + b(X)|X] = a(X)E(Y|X) + b(X)$.

Property CE.3: If X and Y are independent, then $E[Y|X] = E(Y)$

Property CE.4: $E[E(Y|X)] = E(Y)$